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Project Title	Logistic Regression from Scratch for Heart Disease Prediction
Week	Week 1

WEEK-1 Technical Report

Task Name

Logistic Regression from Scratch for Heart Disease Prediction

Abstract

Heart disease is one of the leading causes of death worldwide, making early diagnosis essential for improving patient outcomes. This project implements the Logistic Regression algorithm from scratch using Python, NumPy, and mathematical optimization techniques without relying on machine learning libraries. The model predicts whether a patient is at risk of heart disease based on medical attributes such as age, blood pressure, cholesterol level, and heart rate. By developing the algorithm manually, the project provides a deeper understanding of binary classification, optimization, and predictive analytics in healthcare.

Introduction

Machine learning plays an important role in healthcare by assisting doctors in predicting diseases at an early stage. Logistic Regression is a supervised learning algorithm widely used for binary classification problems. Unlike linear regression, it predicts probabilities using the sigmoid function. This project focuses on implementing Logistic Regression from scratch to classify patients into healthy or heart disease categories while understanding the mathematical principles behind the algorithm.

Theory

The Logistic Regression model is based on the following concepts:

- Binary classification using probability.
- Sigmoid (Logistic) function transforms linear output into probabilities.

- Maximum Likelihood Estimation (MLE) estimates model parameters.
- Gradient Descent minimizes the cost function.
- Decision boundary separates different classes.

Objectives

- Implement Logistic Regression from scratch.
- Build the sigmoid activation function.
- Implement gradient descent optimization.
- Train the model using healthcare data.
- Predict heart disease risk.
- Evaluate model performance using classification metrics.

Methodology

Step 1: Data Collection

Collect patient medical records including age, cholesterol, blood pressure, maximum heart rate, etc.

Step 2: Data Preprocessing

Handle missing values, normalize features, and split data into training and testing sets.

Step 3: Model Development

Implement the sigmoid function, cost function, and gradient descent using NumPy.

Step 4: Prediction

Predict whether a patient has heart disease based on learned model parameters.

Step 5: Evaluation

Measure model accuracy using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

Dataset

The dataset contains:

- Age
- Gender
- Blood Pressure
- Cholesterol
- Maximum Heart Rate
- Chest Pain Type
- Blood Sugar
- Heart Disease (Target Variable)

Mathematical Model

The Logistic Regression model estimates:

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(w^T x + b)}}$$

where:

- **w** = Weight vector
- **b** = Bias
- **x** = Input features

Complexity Analysis

- Training Complexity: **O(n × m × iterations)**
- Prediction Complexity: **O(n × m)**
- Memory Complexity: **O(m)**

where **n** is the number of samples and **m** is the number of features.

Results

The model successfully:

- Predicts heart disease risk.
- Learns decision boundaries using gradient descent.
- Produces reliable binary classification results.
- Supports early healthcare diagnosis.

Applications

- Medical Diagnosis
- Clinical Decision Support
- Healthcare Analytics
- Disease Risk Prediction
- Hospital Management Systems
- Preventive Healthcare

Conclusion

This project demonstrates a complete implementation of Logistic Regression from scratch for heart disease prediction. By manually implementing the sigmoid function, cost function, and gradient descent algorithm, the model provides a clear understanding of binary classification while delivering accurate predictions for healthcare applications.